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1 The infinite spine case

Throughout this section, G denotes an ordered abelian group. For $n \geq 2$ and $a \notin nG$, recall that $\mathfrak{s}_n(a)$ is the largest convex subgroup $H \subseteq G$ such that $a \notin H + nG$. If $a \in nG$, we set $\mathfrak{s}_n(a) = \emptyset$, and we regard $\emptyset$ as being smaller than every convex subgroup. We set $\mathcal{S}_n = \{\mathfrak{s}_n(a) \mid a \in G\}$. The subgroups $\mathfrak{s}_n(a)$ are uniformly definable in a [1, Lemma 2.1]. Consequently, comparisons and equalities between values of the maps $\mathfrak{s}_n$ are formulas in L_{oag} .

Lemma 1.1 ([4, Lemma 2.8(2) and Remark 2.6]). *Let p be a prime, $r \geq 1$, and $a, b \in G$.*

1. *If $0 \leq j < r$, then*

$$\mathfrak{s}_{p^r}(p^j a) = \mathfrak{s}_{p^{r-j}}(a).$$

If $j \geq r$, then $\mathfrak{s}_{p^r}(p^j a) = \emptyset$.

2. *If $\mathfrak{s}_{p^r}(a) \neq \mathfrak{s}_{p^r}(b)$, then*

$$\mathfrak{s}_{p^r}(a + b) = \max\{\mathfrak{s}_{p^r}(a), \mathfrak{s}_{p^r}(b)\}.$$

Lemma 1.2. *Let p be a prime and let $H < H'$ be nonempty elements of $\mathcal{S}_p$. There exists $a \in H'$ such that*

$$\mathfrak{s}_p(a) = H \quad \text{and} \quad \mathfrak{s}_{p^r}(a) < H'$$

for every $r \geq 1$.

Proof. Choose $b \in G$ such that $\mathfrak{s}_p(b) = H$. Since $H' > H$, the maximality of H gives $b \in H' + pG$. We may therefore write $b = a + pc$ with $a \in H'$ and $c \in G$. As $a - b \in pG$, we have $\mathfrak{s}_p(a) = \mathfrak{s}_p(b) = H$.

Fix $r \geq 1$. If $\mathfrak{s}_{p^r}(a) = \emptyset$, there is nothing to prove. Otherwise, put $K = \mathfrak{s}_{p^r}(a)$. If $K \geq H'$, then $a \in H' \subseteq K$, contrary to $a \notin K + p^r G$. Hence, $K < H'$. ■

Lemma 1.3 (Finite coding). *Let p be a prime and*

$$K_0 < K_1 < \cdots < K_N$$

be nonempty elements of $\mathcal{S}_p$. There exists $x \in G$ such that

$$\mathfrak{s}_{p^{i+1}}(x) = K_i$$

for every $0 \leq i \leq N$.

Proof. For each $j < N$, use Lemma 1.2 to choose $a_j \in G$ such that

$$\mathfrak{s}_p(a_j) = K_j \quad \text{and} \quad \mathfrak{s}_{p^r}(a_j) < K_{j+1}$$

for every $r \geq 1$. Also choose $a_N \in G$ with $\mathfrak{s}_p(a_N) = K_N$, and set

$$x = a_0 + pa_1 + \cdots + p^N a_N.$$

Fix $0 \leq i \leq N$. By Lemma 1.1,

$$\mathfrak{s}_{p^{i+1}}(p^i a_i) = \mathfrak{s}_p(a_i) = K_i.$$

If $j < i$, then

$$\mathfrak{s}_{p^{i+1}}(p^j a_j) = \mathfrak{s}_{p^{i+1-j}}(a_j) < K_{j+1} \leq K_i,$$

while, if $j > i$, then $\mathfrak{s}_{p^{i+1}}(p^j a_j) = \emptyset$. Thus, among the summands of x , the unique maximal $\mathfrak{s}_{p^{i+1}}$ -value is K_i . Repeatedly applying Lemma 1.1(2) gives $\mathfrak{s}_{p^{i+1}}(x) = K_i$. $\blacksquare$

Theorem 1.4. *If $\mathcal{S}_p$ is infinite for some prime p , then the dp-rank of G is $\aleph_0$.*

Proof. We work in a sufficiently saturated monster model of $\text{Th}(G)$. Since $\mathcal{S}_p$ is an infinite interpretable linear order, compactness allows us to choose elements

$$(b_{i,k})_{i,k < \omega}$$

such that all the values

$$H_{i,k} := \mathfrak{s}_p(b_{i,k})$$

are nonempty and

$$H_{i,k} < H_{j,l} \quad \text{whenever} \quad i < j \text{ or } (i = j \text{ and } k < l).$$

Indeed, every finite part of these requirements merely asks for a finite chain of distinct nonempty elements of the infinite linear order $\mathcal{S}_p$.

For a function $\eta : \omega \rightarrow \omega$, consider the partial type

$$\Sigma_\eta(x) = \{ \mathfrak{s}_{p^{i+1}}(x) = H_{i,\eta(i)} \mid i < \omega \}.$$

This type is finitely satisfiable. Indeed, given $N < \omega$, the values

$$H_{0,\eta(0)} < H_{1,\eta(1)} < \cdots < H_{N,\eta(N)}$$

form a finite increasing chain, so Lemma 1.3 realizes the first $N + 1$ conditions of Σ_η . By saturation, there is $a_\eta \in G$ realizing Σ_η .

For every $i < \omega$, let

$$\varphi_i(x; y) \text{ be the formula } \mathfrak{s}_{p^{i+1}}(x) = \mathfrak{s}_p(y).$$

As noted above, this is an L_{oag} -formula. For all $\eta : \omega \rightarrow \omega$, $i < \omega$, and $k < \omega$, we have

$$\models \varphi_i(a_\eta; b_{i,k}) \iff H_{i,\eta(i)} = H_{i,k} \iff \eta(i) = k.$$

Hence, these formulas and parameters form an ICT-pattern of depth $\aleph_0$, and therefore the dp-rank of G is at least $\aleph_0$.

Every ordered abelian group has NIP [3]. Since L_{oag} is countable, this gives dp-rank at most $\aleph_0$. Consequently, the dp-rank of G is exactly $\aleph_0$. $\blacksquare$

2 The lower bound in the finite-spine case

We now assume that G is nontrivial and has finite spines. For a prime p and $H \in \mathcal{S}_p \setminus \{\emptyset\}$, recall that

$$H^{[p]} = \bigcap_{K \in G, K > H} (K + pG)$$

and

$$R_H^{\mathfrak{s}_p} = H^{[p]} / (H + pG)$$

by [4, Definition 2.10 and Lemma 2.14(1)]. We also use $\mathcal{S}_{p^r} = \mathcal{S}_p$ for every $r \geq 1$, as established in [4, Lemma 2.9(3)]. We put

$$\mathcal{S}_p^\infty = \{H \in \mathcal{S}_p \setminus \{\emptyset\} \mid |R_H^{\mathfrak{s}_p}| = \infty\}.$$

Lemma 2.1. *Let p be a prime and $H \in \mathcal{S}_p^\infty$. There is a sequence $(a^k)_{k < \omega}$ such that*

$$\mathfrak{s}_p(a^k) = \mathfrak{s}_p(a^k - a^l) = H$$

whenever $k \neq l$. If $H < K$ for some $K \in \mathcal{S}_p \setminus \{\emptyset\}$, the sequence may moreover be chosen inside K .

Proof. Choose elements $(u^k)_{k < \omega}$ whose images in

$$R_H^{\mathfrak{s}_p} = H^{[p]} / (H + pG)$$

are nonzero and pairwise distinct. Thus, neither u^k nor $u^k - u^l$ belongs to $H + pG$, while all these elements belong to $H^{[p]}$. By [4, Lemma 2.14(1)], this says precisely that

$$\mathfrak{s}_p(u^k) = \mathfrak{s}_p(u^k - u^l) = H$$

for $k \neq l$.

Suppose that $H < K$. Since $u^k \in H^{[p]} \subseteq K + pG$, we can write $u^k = a^k + pc_k$ with $a^k \in K$. Replacing u^k by a^k does not change its image in $R_H^{\mathfrak{s}_p}$. The images therefore remain nonzero and pairwise distinct, so the required equalities remain true. $\blacksquare$

Lemma 2.2 (Simultaneous coding for one prime). *Let p be a prime and let*

$$H_0 < H_1 < \cdots < H_{n-1}$$

be elements of $\mathcal{S}_p^\infty$. For each $i < n$, there is a sequence $(a_i^k)_{k < \omega}$ as in Lemma 2.1, and these sequences can be chosen so that $a_i^k \in H_{i+1}$ whenever $i < n-1$. For $\zeta : n \rightarrow \omega$, set

$$b_\zeta = a_0^{\zeta(0)} + pa_1^{\zeta(1)} + \cdots + p^{n-1}a_{n-1}^{\zeta(n-1)}.$$

Then, for every $i < n$,

1. $\mathfrak{s}_{p^{i+1}}(b_\zeta) = H_i$;
2. the image of b_ζ in $R_{H_i}^{\mathfrak{s}_{p^{i+1}}}$ is the same as that of $p^i a_i^{\zeta(i)}$;
3. for every $k < \omega$,

$$b_\zeta - p^i a_i^k \in H_i + p^{i+1}G \iff k = \zeta(i).$$

Proof. The required sequences are supplied by Lemma 2.1. For $j < n-1$, the fact that $a_j^k \in H_{j+1}$ implies

$$\mathfrak{s}_{p^r}(a_j^k) < H_{j+1}$$

for every $r \geq 1$: otherwise, $\mathfrak{s}_{p^r}(a_j^k) \geq H_{j+1}$ would contain a_j^k , contrary to the definition of $\mathfrak{s}_{p^r}$. The calculation in the proof of Lemma 1.3 now gives

$$\mathfrak{s}_{p^{i+1}}(b_\zeta) = H_i,$$

which proves (1).

For $j < i$, we have $a_j^{\zeta(j)} \in H_{j+1} \subseteq H_i$, whereas for $j > i$, the term $p^j a_j^{\zeta(j)}$ belongs to $p^{i+1}G$. Hence,

$$b_\zeta - p^i a_i^{\zeta(i)} \in H_i + p^{i+1}G.$$

Also, by Lemma 1.1,

$$\mathfrak{s}_{p^{i+1}}(p^i a_i^{\zeta(i)}) = \mathfrak{s}_p(a_i^{\zeta(i)}) = H_i.$$

Thus, both elements represent classes in

$$R_{H_i}^{\mathfrak{s}_{p^{i+1}}} = H_i^{[p^{i+1}]} / (H_i + p^{i+1}G),$$

and the congruence $b_\zeta - p^i a_i^{\zeta(i)} \in H_i + p^{i+1}G$ proves (2).

If $k \neq \zeta(i)$, then

$$p^i (a_i^{\zeta(i)} - a_i^k) \notin H_i + p^{i+1}G.$$

Indeed, by the purity of the convex subgroup H_i , membership in the group on the right would imply

$$a_i^{\zeta(i)} - a_i^k \in H_i + pG,$$

contrary to Lemma 2.1. Together with the preceding congruence, this proves (3). $\blacksquare$

The following standard consequences of compactness will be used without proof. For the second assertion, finite satisfiability follows from the Chinese remainder theorem [4, Lemma 2.1(2)].

Lemma 2.3. *Let G be a sufficiently saturated monster model of the theory of a nontrivial ordered abelian group.*

1. *The type-definable subgroup*

$$D = \bigcap_{m \geq 2} mG$$

is cofinal and coinital in G .

2. *Let $P_0 \subseteq \mathbb{P}$ and $(a_p)_{p \in P_0}$ be a sequence from G . There exists $a \in G$ such that*

$$a - a_p \in p^r G$$

for every $p \in P_0$ and $r \geq 1$.

Theorem 2.4. *Let G be a nontrivial ordered abelian group with finite spines. Then*

$$\text{dp-rk}(G) \geq 1 + \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|.$$

If the sum on the right is infinite, then the dp-rank of G is $\aleph_0$.

Proof. We work in a sufficiently saturated monster model. For each prime p , put $n_p = |\mathcal{S}_p^\infty|$ and, when $n_p > 0$, enumerate

$$\mathcal{S}_p^\infty = \{H_{p,0} < \cdots < H_{p,n_p-1}\}.$$

The number n_p is finite because G has finite spines. Let

$$\mathcal{I} = \{(p, i) \mid p \in \mathbb{P}, 0 \leq i < n_p\}.$$

Thus,

$$|\mathcal{I}| = \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|,$$

and $\mathcal{I}$ is finite or countable.

For each p with $n_p > 0$, choose sequences $(a_{p,i}^k)_{k < \omega}$ as in Lemma 2.2, and denote the corresponding coding elements by $b_{p,\zeta}$ for $\zeta : n_p \rightarrow \omega$. For $(p, i) \in \mathcal{I}$, let

$$\varphi_{p,i}(x; y) \quad \text{be the formula} \quad x - y \in \mathfrak{s}_{p^{i+1}}(y) + p^{i+1}G.$$

This is an L_{oag} -formula by the uniform definability of $\mathfrak{s}_{p^{i+1}}$. We use the parameter sequence

$$(p^i a_{p,i}^k)_{k < \omega}$$

in the row indexed by (p, i) . By Lemma 1.1,

$$\mathfrak{s}_{p^{i+1}}(p^i a_{p,i}^k) = H_{p,i}.$$

Fix a function $\eta : \mathcal{I} \rightarrow \omega$ and consider

$$q_\eta(x) = \left\{ \varphi_{p,i}(x; p^i a_{p,i}^k)^{\eta(p,i)=k} \mid (p,i) \in \mathcal{I}, k < \omega \right\}.$$

Here, the superscript means that the formula is negated when $\eta(p,i) \neq k$. For each p with $n_p > 0$, let

$$\eta_p : n_p \rightarrow \omega, \quad \eta_p(i) = \eta(p,i).$$

By Lemma 2.3(2), there is $b_\eta \in G$ such that

$$b_\eta - b_{p,\eta_p} \in p^r G$$

for every p with $n_p > 0$ and $r \geq 1$. Part (3) of Lemma 2.2 then shows that $b_\eta \models q_\eta$.

It remains to add the row arising from the order. Since the monster is sufficiently saturated, there exists $c > 0$ such that

$$-c < b_\eta < c$$

for every $\eta : \mathcal{I} \rightarrow \omega$. By Lemma 2.3, choose

$$t_0 < t_1 < \dots$$

from D so that $t_{k+1} - t_k > 2c$ for every $k < \omega$. Set

$$I_k = (t_k - c, t_k + c).$$

These intervals are pairwise disjoint and satisfy $I_0 < I_1 < \dots$. We take

$$\psi(x; y, z) := (y < x < z)$$

for the additional row, with parameter sequence

$$(t_k - c, t_k + c)_{k < \omega}.$$

Let $\nu : \{*\} \dot{\cup} \mathcal{I} \rightarrow \omega$, and put $\eta = \nu|_{\mathcal{I}}$. The element

$$a_\nu = b_\eta + t_{\nu(*)}$$

belongs to $I_{\nu(*)}$. Moreover, $t_{\nu(*)} \in p^{i+1}G$ for every $(p,i) \in \mathcal{I}$, so translation by $t_{\nu(*)}$ does not change the truth of any formula $\varphi_{p,i}$. It follows that

$$\models \psi(a_\nu; t_k - c, t_k + c) \iff \nu(*) = k$$

and

$$\models \varphi_{p,i}(a_\nu; p^i a_{p,i}^k) \iff \nu(p,i) = k.$$

We have therefore constructed an ICT-pattern of depth

$$1 + |\mathcal{I}| = 1 + \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|,$$

as required. If the sum is infinite, the ICT-pattern has depth $\aleph_0$. The NIP upper bound recalled in the proof of Theorem 1.4 gives $\text{dp-rk}(G) \leq \aleph_0$, so equality holds in this case. $\blacksquare$

3 The upper bound in the finite-spine case

We continue to assume that G is nontrivial and has finite spines, and now also assume that

$$N = \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|$$

is finite. We use the following standard criterion.

Fact 3.1 (Indiscernible-sequence criterion). *Let T be a complete theory and let $n \geq 1$. It is enough to consider sequences indexed by a fixed model of DLO; we take this model I to be Dedekind complete. Suppose that, whenever $(\bar{a}_i)_{i \in I}$ is an indiscernible sequence and b is a singleton, one can divide I by at most n points so that*

$$\text{tp}(\bar{a}_i/b) = \text{tp}(\bar{a}_j/b)$$

whenever i and j lie in the same convex component of the remainder. Then $\text{dp-rk}(T) \leq n$.

This may be proved in the same way as [2, Fact 2.10].

Henceforth I denotes this Dedekind complete index set.

We will also use the following specialization of relative quantifier elimination.

Proposition 3.2 ([4, Proposition 2.26]). *Let G be an ordered abelian group with finite spines and let f be a partial map from G to G whose domain is a subgroup. Suppose that f is an ordered group embedding and, for every a in its domain, every prime p , and every $r \geq 1$, the following hold:*

$$\begin{aligned} \mathfrak{s}_{p^r}(f(a)) &= \mathfrak{s}_{p^r}(a), & \mathfrak{t}_p(f(a)) &= \mathfrak{t}_p(a), \\ a =_{\bullet} k_{\bullet} &\iff f(a) =_{\bullet} k_{\bullet}, & a \equiv_{p^r, \bullet} k_{\bullet} &\iff f(a) \equiv_{p^r, \bullet} k_{\bullet}. \end{aligned}$$

Here the third condition is required for every $k \in \mathbb{Z} \setminus \{0\}$, and the fourth for every $1 \leq k < p^r$. Then f is elementary.

Proof. We use the identity map on all the auxiliary sorts in the cited proposition. The reduction of the predicates $x \equiv_{m, \bullet} k_{\bullet}$ to prime-power moduli is given by [4, Lemma 2.22]. It only remains to explain why its predicates $D_{p^r}^{[p^s]}$, where $s \geq r$, require no additional hypothesis.

Indeed, in the finite-spine case,

$$D_{p^r}^{[p^s]}(a) \iff a \notin p^r G.$$

The forward implication is immediate from the definition. Conversely, put $H = \mathfrak{s}_{p^r}(a)$. If H has a successor H' in $\mathcal{S}_p$, then

$$H^{[p^s]} = H' + p^s G,$$

while, if H has no successor, then $H^{[p^s]} = G$. In either case, the definition of $\mathfrak{s}_{p^r}(a)$ gives

$$a \in H^{[p^s]} + p^r G \quad \text{and} \quad a \notin H + p^r G.$$

Thus H witnesses $D_{p^r}^{[p^s]}(a)$. Since preservation of $\mathfrak{s}_{p^r}$ also preserves membership in $p^r G$, the result follows. $\blacksquare$

Lemma 3.3 ([4, Lemma 3.4]). *Let p be a prime, $H \in \mathcal{S}_p \setminus \{\emptyset\}$, and $r \geq 1$. If $R_H^{s_p}$ is finite, then $R_H^{s_{p^r}}$ is finite. Consequently,*

$$|R_H^{s_{p^r}}| = \infty \implies H \in \mathcal{S}_p^\infty.$$

The next lemma isolates the single contribution coming from the order and from the ordered ribs associated with the maps $\mathfrak{t}_p$.

Lemma 3.4 (The common order cut). *Let $(\bar{a}_i)_{i \in I}$ be indiscernible, and let $b \in G$. One can divide I by at most one point so that the following data are constant on each convex component of the remainder:*

1. the sign of $\bar{m} \cdot \bar{a}_i - mb$;
2. $\mathfrak{t}_p(\bar{m} \cdot \bar{a}_i - mb)$;
3. the truth value of

$$\bar{m} \cdot \bar{a}_i - mb =_{\bullet} k_{\bullet}.$$

Here the assertions hold simultaneously for all integer tuples $\bar{m}$, all nonzero integers m , all primes p , and all nonzero integers k .

Proof. Let $\mathcal{H}$ consist of $\{0\}$ and all nonempty elements of the finite sorts $\mathcal{T}_p$. (These coincide with the sorts $\mathcal{S}_p$ in the finite-spine case, immediately from the definitions.) Every member of $\mathcal{H}$ is algebraic as an imaginary over $\emptyset$, so an infinite indiscernible sequence remains indiscernible over $\text{acl}^{\text{eq}}(\emptyset)$.

Fix $H \in \mathcal{H}$, an integer tuple $\bar{m}$, a nonzero integer m , and an integer k , where either $k = 0$ or G/H is discrete. The element

$$c_i = (\bar{m} \cdot \bar{a}_i)/H - k_H \in G/H$$

is obtained from $\bar{a}_i$ by a map defined over H , hence over $\text{acl}^{\text{eq}}(\emptyset)$. Thus $(c_i)_{i \in I}$ may be treated as part of the indiscernible sequence. In particular, it is either constant or strictly monotone. Consequently, writing $x_i = c_i - m(b/H)$, exactly one of the following alternatives holds:

- (i) the sign of x_i is constant as i varies in I ;
- (ii) there exists $i_0 \in I$ such that the terms x_i for $i < i_0$ are all negative and the terms x_i for $i > i_0$ are all positive, or vice versa.

(The realization of the cut as a point of I uses Dedekind completeness; at most one index can satisfy $x_i = 0$.)

Now suppose that (ii) holds at i_0 , and let $K \in \mathcal{H}$ satisfy $K > H$. Let $\pi: G/H \rightarrow G/K$ be the canonical projection. The constant k_H maps to zero in G/K : if G/H is discrete, every nonzero convex subgroup of G/H contains its least positive element. Hence $\pi(x_i)$ may be identified with $c_i/K - m(b/K)$. The sequence $(\pi(x_i))_{i \in I}$ is constant or strictly monotone. Assume, without loss of generality, that $x_i < 0$ for $i < i_0$ and $x_i > 0$ for $i > i_0$; the opposite orientation is symmetric. Then $\pi(x_i) \leq 0 \leq \pi(x_j)$ whenever $i < i_0 < j$. If

$(\pi(x_i))_{i \in I}$ is constant, it is therefore the zero sequence, and (i) holds in G/K . If it is strictly monotone, then it is strictly increasing, and $\pi(x_i) = 0$ is impossible for $i < i_0$: otherwise density of I would furnish i' with $i < i' < i_0$, whence $0 = \pi(x_i) < \pi(x_{i'}) \leq 0$. Thus $\pi(x_i) < 0$ for all $i < i_0$, and likewise $\pi(x_j) > 0$ for all $j > i_0$. This is (ii) at the same point i_0 .

We next show that distinct points of I cannot arise in alternative (ii). Suppose toward a contradiction that configurations associated with subgroups H and K do so at points $\gamma < \delta$ of I . Write $c_i \in G/H$ and $d_i \in G/K$ for the corresponding sequences, with coefficients m and m' of b . We may assume $H \leq K$. Replacing (c_i) by $m'c_i$ and (d_i) by md_i leaves the exceptional points unchanged, so we may assume $m = m'$. Replacing both sequences by their negatives if necessary (and changing the common coefficient of b accordingly), we may assume that $(c_i)_{i \in I}$ is nondecreasing. Choose

$$h < \gamma < r < s < \delta < l$$

distinct from the at most two indices at which either sequence may vanish.

Since $H \leq K$, it is enough to project the sequence associated with H . If $H < K$, the preceding paragraph yields that $(c_i - m(b/K))_{i \in I}$ satisfies (i) or else (ii) at γ ; if $H = K$, this sequence is the original one and already satisfies (ii) at γ . The sequence $(d_i - m(b/K))_{i \in I}$ lives in G/K and satisfies (ii) at δ .

Suppose first that the (possibly projected) H -sequence still satisfies (ii) at γ . Then (c_i) and (d_i) are monotone sequences in G/K crossing the same element $m(b/K)$ at the distinct points γ and δ . If (d_i) is also nondecreasing, the earlier crossing may be that of (c_i) or that of (d_i) . In the former case one has $c_r > d_s$ but $c_h < d_l$; in the latter case, $d_r > c_s$ but $d_h < c_l$. Either pair of inequalities contradicts indiscernibility, since both (r, s) and (h, l) are increasing pairs. If instead (d_i) is decreasing, then $c_i < d_i$ holds for all $i < \gamma$ and fails for all $i > \delta$, again contradicting indiscernibility.

It remains to treat the case where the projected H -sequence satisfies (i). Then $H < K$ and $c_i = m(b/K)$ for every i , so

$$d_i - m(b/K) = d_i - c_j$$

holds in G/K for every j . Choosing $j < i$, the sign of the right-hand side is independent of the increasing pair (j, i) by indiscernibility. Hence the K -sequence cannot satisfy (ii), a contradiction.

Thus at most one point of I arises in alternative (ii). Dividing I by this point if it exists, the signs under consideration are constant on each convex component of the remainder. Taking $H = \{0\}$ and $k = 0$ gives (1). For (2), the value $t_p(x)$ is determined by the truth values of $x \in H$ for $H \in \mathcal{T}_p$, and these are determined by the signs of x/H . Finally, by [4, Lemma 2.21(1)], a witness to $x = \bullet k \bullet$ is $t_2(x)$. Thus (2), together with constancy of the sign of $x/H - k_H$, gives (3). ■

We now control all contributions from the $\mathfrak{s}_{pr}$ -side. Let $(\bar{a}_i)_{i \in I}$ and b be as in Lemma 3.4. For a prime p and for H equal to $\{0\}$ or a nonempty element of $\mathcal{S}_p$, call $i_0 \in I$ a (p, H) -exception if there are an integer tuple $\bar{m}$, a nonzero

integer m , an integer $r \geq 1$, and an integer k such that either $k = 0$ or G/H is discrete, and such that, writing

$$c_i = (\overline{m} \cdot \overline{a}_i)/H - k_H \in G/H,$$

one has

$$c_i \equiv_{p^r} m(b/H) \iff i = i_0.$$

Lemma 3.5 (Uniqueness of the congruence exception). *Let p be a prime and let H be $\{0\}$ or a nonempty element of $\mathcal{S}_p$.*

1. *There is at most one (p, H) -exception.*
2. *Suppose i_0 is a (p, H) -exception, witnessed by $\overline{m}$, m , r , and k as above, and set*

$$K = \mathfrak{s}_{p^r}(\overline{m} \cdot \overline{a}_j - \overline{m} \cdot \overline{a}_{i_0})$$

for $i \neq j$ in I (this value is independent of the pair). Then $K \in \mathcal{S}_p^\infty$ and i_0 is a (p, K) -exception.

Proof. For (1), suppose that i_0 and i_1 are exceptions, witnessed respectively by

$$(c_i, m, r) \quad \text{and} \quad (d_i, m', r')$$

in the notation above. In G/H one has, for every $n \geq 1$,

$$x \equiv_{p^n} 0 \iff p^s x \equiv_{p^{n+s}} 0$$

for all $s \geq 0$, by purity of H , and

$$x \equiv_{p^n} 0 \iff sx \equiv_{p^n} 0$$

whenever s is coprime to p .

Assume without loss of generality that $r \geq r'$. The first equivalence, applied with exponent $r - r'$, replaces the second congruence by an equivalent congruence modulo p^r . The second equivalence lets us multiply by an inverse of the prime-to- p part of each coefficient of b , without changing the solution sets. The sequences remain of the same shape, and the exceptional congruences may thus be written

$$\begin{aligned} c_i \equiv_{p^r} p^e(b/H) &\iff i = i_0, \\ d_i \equiv_{p^r} p^f(b/H) &\iff i = i_1, \end{aligned}$$

for some integers $e, f \geq 0$.

Assume first that $i_0 < i_1$. If $e \leq f$, eliminating b gives

$$d_{i_1} - p^{f-e} c_{i_0} \equiv_{p^r} 0.$$

By indiscernibility over the algebraic imaginary H , the same relation holds with i_1 replaced by any $j > i_0$. Subtracting yields $d_j \equiv_{p^r} d_{i_1}$, so the second exceptional congruence also holds at j , a contradiction. If $e > f$, eliminate b

in the other direction and replace the increasing pair (i_0, i_1) by (h, i_1) for some $h < i_1$ with $h \neq i_0$. This gives $c_h \equiv_{p^r} c_{i_0}$, contradicting uniqueness of the first congruence. The case $i_1 < i_0$ is symmetric.

For (2), independence of the pair follows by indiscernibility. Write $x_i = \bar{m} \cdot \bar{a}_i - mb$, so that $K = \mathfrak{s}_{p^r}(x_j - x_i)$. The congruence $c_i \equiv_{p^r} m(b/H)$ holds at exactly one index, so the c_i are not all congruent modulo p^r . Thus $K \neq \emptyset$ and $H \leq K$.

If $K = H$, then $\mathfrak{s}_{p^r}(x_j - x_i) = H$ for $i \neq j$, so the images of these differences in $R_H^{\mathfrak{s}_{p^r}}$ are pairwise distinct and nonzero. Hence $H \in \mathcal{S}_p^\infty$ by Lemma 3.3, and i_0 is already a (p, K) -exception.

Now suppose $H < K$. If $k = 0$, then $x_{i_0} \in H + p^r G \subseteq K + p^r G$. If $k \neq 0$, then G/H is discrete, so k_H maps to zero in G/K , and the congruence at i_0 projects to $x_{i_0}/K \equiv_{p^r} 0$. In either subcase, $\mathfrak{s}_{p^r}(x_{i_0}) < K = \mathfrak{s}_{p^r}(x_i - x_{i_0})$ for $i \neq i_0$, whence $\mathfrak{s}_{p^r}(x_i) = K$ for $i \neq i_0$ by Lemma 1.1(2). Thus $x_i \in K + p^r G$ if and only if $i = i_0$, so i_0 is a (p, K) -exception with $k = 0$. The images of the x_i for $i \neq i_0$ in $R_K^{\mathfrak{s}_{p^r}}$ are pairwise distinct, hence $K \in \mathcal{S}_p^\infty$ by Lemma 3.3. ■

Lemma 3.6 (Control of the $\mathfrak{s}$ -data). *For every pair (p, H) with $H \in \mathcal{S}_p^\infty$, choose its exceptional index if one exists, and let E be the set of indices so obtained. Then $|E| \leq N$. If $C \subseteq I$ is convex and $C \cap E = \emptyset$, then for any $i, j \in C$, every integer tuple $\bar{m}$, every nonzero integer m , every prime p , every $r \geq 1$, and every $1 \leq k < p^r$, we have*

$$\begin{aligned} \mathfrak{s}_{p^r}(\bar{m} \cdot \bar{a}_i - mb) &= \mathfrak{s}_{p^r}(\bar{m} \cdot \bar{a}_j - mb), \\ \bar{m} \cdot \bar{a}_i - mb &\equiv_{p^r, \bullet} k_\bullet \iff \bar{m} \cdot \bar{a}_j - mb \equiv_{p^r, \bullet} k_\bullet. \end{aligned}$$

Proof. By Lemma 3.5(1) one has $|E| \leq N$. By part (2) of the same lemma, every (p, H) -exception already arises as a (p, K) -exception for some $K \in \mathcal{S}_p^\infty$, hence belongs to E .

Fix an integer tuple $\bar{m}$, a nonzero integer m , a prime p , and $r \geq 1$, and set

$$x_i = \bar{m} \cdot \bar{a}_i - mb.$$

Let H be $\{0\}$ or a nonempty element of $\mathcal{S}_p$, and let k be 0 or such that G/H is discrete. Put

$$c_i = (\bar{m} \cdot \bar{a}_i)/H - k_H \in G/H.$$

By indiscernibility over H , either $c_i \equiv_{p^r} c_j$ for all distinct $i, j \in I$, or else $c_i \not\equiv_{p^r} c_j$ for all distinct $i, j \in I$. In the first case, $c_i - m(b/H) \equiv_{p^r} c_j - m(b/H)$ for all i, j , so the condition $x_i/H \equiv_{p^r} k_H$ is independent of i . In the second case, there is at most one $i_0 \in I$ such that $c_{i_0} - m(b/H) \equiv_{p^r} 0$. If this i_0 exists, then i_0 is a (p, H) -exception, so $i_0 \in E$.

Let $C \subseteq I$ be convex and disjoint from E . Then $x_i/H \equiv_{p^r} k_H$ is independent of $i \in C$. Applying this with $k = 0$ as H varies, we obtain that $\mathfrak{s}_{p^r}(x_i)$ is constant on C . If the common value is $\emptyset$, the predicate $x_i \equiv_{p^r, \bullet} k_\bullet$ fails on C . Otherwise, by [4, Lemma 2.21(2)], a witness to the predicate is this common value H . If G/H is dense, the predicate fails on C ; if G/H is discrete, constancy of $x_i/H \equiv_{p^r} k_H$ on C is the case already treated. ■

Theorem 3.7. *Let G be a nontrivial ordered abelian group with finite spines, and suppose that*

$$\sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|$$

is finite. Then

$$\text{dp-rk}(G) \leq 1 + \sum_{p \in \mathbb{P}} |\mathcal{S}_p^\infty|.$$

Proof. We work in a sufficiently saturated monster model. Let $(\bar{a}_i)_{i \in I}$ be indiscernible, and let $b \in G$. Apply Lemma 3.4 and Lemma 3.6. Dividing I by the common order-cut point together with the at most N points of E produces at most $N + 1$ points.

Fix a convex component C of the remainder, and let $i, j \in C$. Let M_i be the subgroup generated by the entries of $\bar{a}_i$ together with b , and define

$$f_{ij} : M_i \longrightarrow M_j$$

by

$$f_{ij}(\bar{m} \cdot \bar{a}_i + mb) = \bar{m} \cdot \bar{a}_j + mb.$$

The case $m = 0$ in the following verifications follows directly from indiscernibility over $\text{acl}^{\text{eq}}(\emptyset)$; in particular, the actual values in the finite auxiliary sorts, and not merely their types, agree. For $m \neq 0$, Lemma 3.4(1) shows that the sign of every element is preserved. If two displayed representations of an element of M_i are equal, subtracting them gives an integer-linear expression of sign zero. Its corresponding expression at j also has sign zero, so f_{ij} is well-defined. The same argument for nonzero signs gives injectivity and order preservation. Thus f_{ij} is an ordered group embedding. The same lemma, parts (2) and (3), shows that f_{ij} preserves every $\mathfrak{t}_p$ and every predicate $x =_{\bullet} k_{\bullet}$. By Lemma 3.6, it preserves every $\mathfrak{s}_{p^r}$ and every predicate $x \equiv_{p^r, \bullet} k_{\bullet}$. Therefore, Proposition 3.2 implies that f_{ij} is elementary. Since it fixes b and sends $\bar{a}_i$ to $\bar{a}_j$, we obtain

$$\text{tp}(\bar{a}_i/b) = \text{tp}(\bar{a}_j/b).$$

We have verified the hypothesis of Fact 3.1 with $n = N + 1$. Hence $\text{dp-rk}(G) \leq N + 1$, as required. $\blacksquare$

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